The Role of AI Literacy in Trust Calibration toward AI-Generated Content

Course Research Project — Questionnaire Presentation

Diogo Costa Madalena Martins Maja Skwaron Rafael Gufler

May 13, 2026

Human-AI Interaction Course

Outline

Part 1 – Research Design

- Research question & hypotheses
- Variables and constructs

Part 2 – Theoretical Background

- AI literacy as a construct
- Trust in AI: HAITS

Part 3 – Instrument

- Overview of the questionnaire
- Objective knowledge test (OLS)
- Trust subscales

Part 4 – Analysis & Limitations

- Analysis protocol
- Known limitations

Part 1 – Research Design

Research Question & Hypotheses

Research Question

Does **objective AI literacy** predict **trust in AI tools** among university students?

Research Question

Does **objective AI literacy** predict **trust in AI tools** among university students?

Core argument

Users who understand how AI actually works — probabilistic generation, hallucination tendencies, bias mechanisms, knowledge limits — should be **less likely** to trust its outputs uncritically.

Research Question & Hypotheses

Research Question

Does **objective AI literacy** predict **trust in AI tools** among university students?

Core argument

Users who understand how AI actually works — probabilistic generation, hallucination tendencies, bias mechanisms, knowledge limits — should be **less likely** to trust its outputs uncritically.

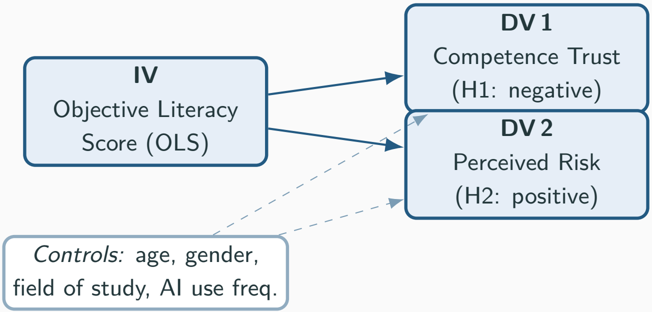
H1 — Competence Trust

Higher AI literacy $\Rightarrow$
lower Competence Trust in AI tools.

H2 — Perceived Risk

Higher AI literacy $\Rightarrow$
higher Perceived Risk when using AI tools.

Variables and Constructs



Exploratory (not hypothesis tests)

Overconfidence gap: self-estimated OLS minus actual OLS. Domain-specific trust: 8 contexts from low- to high-stakes.

Part 2 – Theoretical Background

AI Literacy as a Construct

Key definitions

- **Long & Magerko (2020, CHI):** AI literacy = competencies to critically evaluate, communicate with, and use AI.
- **Ng et al. (2022):** four domains — knowing, using, evaluating, ethical AI.
- This study focuses on **objective** literacy: performance-based knowledge of how AI systems work.

Knowledge test source

Hornberger et al. (2023)
Computers and Education: AI

- Only peer-reviewed, IRT-validated AI literacy test.
- Validated on $N = 1{,}286$ German university students.
- This study: **11 items** translated to English.

Trust in AI: the HAITS

Human-AI Trust Scale — Sun et al. (2025)

Validated on $N = 1{,}426$ (China + US). Four factors: Affective Trust, **Competence Trust**, Benevolence & Integrity, **Perceived Risk**.

Trust in AI: the HAITS

Human-AI Trust Scale — Sun et al. (2025)

Validated on $N = 1{,}426$ (China + US). Four factors: Affective Trust, **Competence Trust**, Benevolence & Integrity, **Perceived Risk**.

Competence Trust (used: H1)

Does the user believe AI is **accurate and reliable**?

Perceived Risk (used: H2)

Does the user believe AI **could cause harm or deception**?

Why not all four subscales?

Affective Trust and Benevolence & Integrity measure *relational bonding* (e.g. “AI is my close friend”) — outside the scope of trust in AI-generated *content*.

Part 3 – Instrument

Questionnaire Overview

5 sections $\approx$ 8–10 minutes. Target: university students.

Section	Content	Items	Role
1	Demographics & background	6	Controls
2	AI usage behaviour	4 + 1	Control + Exploratory
3	Objective AI knowledge test	12	**IV — OLS + Exploratory**
4	Trust in AI tools (HAITS)	11 + 1	**DVs**
5	Domain-specific trust	8	Exploratory

Questionnaire Overview

5 sections ≈ 8–10 minutes. Target: university students.

Section	Content	Items	Role
1	Demographics & background	6	Controls
2	AI usage behaviour	4 + 1	Control + Exploratory
3	Objective AI knowledge test	12	**IV — OLS + Exploratory**
4	Trust in AI tools (HAITS)	11 + 1	**DVs**
5	Domain-specific trust	8	Exploratory

Two embedded attention checks

Item 2.4: "What is 4 + 4? Please type the number." Item 4.12: "Please select 'Agree' for this item."

Responses failing either check are excluded from analysis.

Section 3 — Objective Literacy Score (OLS)

11 knowledge items (Hornberger et al., 2023), grouped into four themes:

1. ML fundamentals
Quality of a model, supervised vs. unsupervised learning, training/test split.

2. Human role in AI
How and to what extent humans influence ML outcomes.

3. How AI works
Goal-directedness, learning from large data, recommender systems.

4. Limitations & ethics
Unrepresentative training data, black-box problem, predictive policing risks.

Scoring

1 point per correct answer; 5 options per item including "I don't know".

OLS range: 0–11. Item 3.12: self-estimate (overconfidence gap = estimate − OLS).

Section 4 — HAITS Trust Items

Competence Trust (4.1–4.6)

1 = Strongly disagree, 5 = Strongly agree

1. The AI tools are reliable.
2. AI tools are competent and perform their tasks well.
3. AI tools perform their role very well.
4. AI tools can do everything I would expect them to.
5. I can trust the information presented to me by AI tools.
6. AI tools are fast and efficient.

Perceived Risk (4.7–4.11) — all reverse-coded (R)

1 = Strongly disagree, 5 = Strongly agree

7. The AI tools are deceptive. (R)
8. The AI tools behave in an underhanded manner. (R)
9. I am suspicious of the AI tools' intent or outputs. (R)
10. The AI tools' actions could lead to harmful outcomes. (R)
11. It is risky to interact with AI tools. (R)

Scoring note

For Perceived Risk, higher scores indicate that students see AI tools as more risky, which reflects lower trust.

Section 5 — Domain-Specific Trust

8 domains rated on: *1 = Not at all, 5 = Completely*

Informed by Nguyen et al. (2025) — AI trust and literacy in student samples.

Lower-stakes domains

- General knowledge questions
- Learning or study-related tasks
- Programming or technical tasks
- Professional or work-related tasks

Higher-stakes domains

- Writing academic work (essays, papers)
- **Health-related questions**
- **Personal advice or life decisions**
- News and current events

Exploratory purpose

If H1 and H2 hold, literacy effects are expected to be strongest in high-stakes domains - where the cost of misplaced trust is highest - and weakest in low-stakes domains.

Part 4 – Analysis & Limitations

Analysis Protocol — Steps 1 & 2

Step 1 — Data preparation

- **Exclude** responses failing either attention check.
- **Exclude** bottom 5th percentile completion times

  (implausibly fast = not reading carefully)
- **Score OLS** (0–11): 1 = correct, 0 = incorrect or "I don't know".
- **Reverse-code** Perceived Risk items 4.7–4.11

  (subtract from 6, so high score = high risk)
- **Overconfidence gap** = item 3.12 − actual OLS

  (positive = overconfident; reported descriptively)

Step 2 — Scale reliability

- Compute **Cronbach's** α and **McDonald's** ω for both subscales.

  α/ω measure whether items in a scale consistently point at the same construct. Target: $\alpha \geq .70$.
- **Benchmarks from Sun et al. (2025):**

  Competence Trust $\omega = .966$

  Perceived Risk $\omega = .959$
- If $\alpha < .70$: inspect item-total correlations; remove weakest item.
- If $N \geq 150$: optional **two-factor CFA** in `lavaan`

  confirms the two-factor structure holds in our sample; report CFI, RMSEA, SRMR

Analysis Protocol — Steps 3 & 4

Step 3 — Descriptives

- Means and SDs for all constructs.
- **Correlation matrix**: check that OLS correlates negatively with Competence Trust and positively with Perceived Risk before running regressions.
- **One-sample *t*-test** on overconfidence gap vs. zero: tests whether students systematically over- or underestimate their own AI knowledge.

Step 4 — Hypothesis testing

Two **multiple regressions**, one per hypothesis.

H1: OLS $\rightarrow$ Competence Trust

Expected: $\beta < 0$

H2: OLS $\rightarrow$ Perceived Risk

Expected: $\beta > 0$

Both models control for age, gender, field of study, and AI use frequency.

Controls remove variance explained by, for example, STEM students having both higher literacy and different trust baselines.

Report: standardized β, R^2, p-value, Cohen's f^2.

$f^2 = .02$ small; .15 medium; .35 large.

Sample Size

Power analysis

- Two multiple regressions with 4 predictors.
- Assumed effect size: $f^2 = .15$ (medium), based on the AI literacy–attitude literature.
- At power $= .80$ and $\alpha = .05$: **minimum** $N \approx 85$.
- **Target:** $N = 100$–150 to allow for exclusions such as attention-check failures or very fast completions.

Known Limitations (Pre-Acknowledged)

1. **No dispositional trust control.** Baseline technology enthusiasm is not measured; field of study and prior AI coursework serve as proxy controls. Future work should include a validated dispositional trust scale (McKnight et al., 2011).
2. **Cross-linguistic transfer of knowledge items.** Hornberger et al.'s test was developed in German; IRT item parameters may not transfer directly to an English-language sample. Item-level descriptives will characterize the scale empirically.
3. **Partial HAITS.** Only two of four subscales used (Competence Trust + Perceived Risk). Affective Trust and Benevolence & Integrity are out of scope but relevant for future research on heavy AI users.
4. **Student sample.** University students have above-average digital literacy; variance in both literacy and trust may be compressed relative to the general population.

Appendix A – Construct Map

Appendix A — Construct Map (Controls & Exploratory)

Items	Construct	Role	Source
1.1–1.2	Age, gender	Control	Brown et al. (2025)
1.3–1.4	Field of study, status	Control	Hornberger et al. (2023)
1.5	AI course experience	Control	KPMG (2025)
1.6	AI usage duration	Control	Developed for study
2.1	AI use frequency	Control	Brown et al. (2025)
2.2	Weekly AI time	Control	Developed for study
2.3 + 2.5	Self-rated confidence	*Exploratory* — overconfidence gap	DSIT (2026); KPMG (2025)
3.1–3.11	Objective AI literacy	**Main IV — H1 & H2**	Hornberger et al. (2023)
3.12	Self-estimated score	*Exploratory* — overconfidence gap	Welsch et al. (2025)
4.1–4.6	Competence Trust	**Primary DV — H1**	Sun et al. (2025) HAITS
4.7–4.11	Perceived Risk (R)	**Primary DV — H2**	Sun et al. (2025) HAITS
5.1–5.8	Domain-specific trust	*Exploratory*	Nguyen et al. (2025)

Appendix B – Key References

Appendix B — Key References

- Brown, R., Sillence, E., & Branley-Bell, D. (2025). AcademAI: Investigating AI usage, attitudes, and literacy in higher education and research. *Journal of Educational Technology Systems*, 54(1), 6–33.
- Department for Science, Innovation and Technology & DCMS. (2026). *AI skills for life and work: General public survey findings*. GOV.UK.
- Hornberger, M., Bewersdorff, A., & Nerdel, C. (2023). What do university students know about Artificial Intelligence? Development and validation of an AI literacy test. *Computers and Education: Artificial Intelligence*, 5, 100165.
- Koch, M. J., et al. (2024). Meta AI literacy scale: Further validation and development of a short version. *Heliyon*, 10(21), e39686.
- KPMG International & University of Melbourne. (2025). *Trust, attitudes and use of artificial intelligence: A global study 2025*.
- Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. *Proceedings of CHI 2020*, 1–16.
- Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3), 709–734.
- Ng, D. T. K., et al. (2022). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, 100041.
- Nguyen, N. B., et al. (2025). Learning with algorithms: AI usage, trust, and literacy among university students. *HCMC Open University Journal of Science*, 16(10).
- Sun, H., et al. (2025). Revisiting trust in the era of generative AI: Factorial structure and latent profiles. arXiv:2510.10199.
- Welsch, R., et al. (2025). AI makes you smarter but none the wiser. *Computers in Human Behavior*, 168, 108262.

Thank you!

Questions?

The Role of AI Literacy in Trust Calibration toward AI-Generated Content
Course Research Project — Questionnaire Presentation